FROM GROTHENDIECK K_0 TO QUILLEN'S HIGHER ALGEBRAIC K-THEORY

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Abstract. These notes isolate the conceptual and technical core of Quillen's higher algebraic K-theory. The emphasis is on the passage from exact categories to the Q-construction, the use of classifying spaces, Quillen's Theorem B, and the four structural results that make the theory calculable: additivity, resolution, dévissage, and localization. The comparison with the $B\mathrm{GL}^+$ construction is also explained in detail. A short post-Quillen outlook is placed in an appendix, as requested.

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1. Historical prelude and the problem Quillen solved

Classical algebraic K-theory begins with Grothendieck's group K_0, then extends to K_1 and K_2 through the work of Bass, Whitehead, Steinberg, and Milnor. What remained genuinely missing around 1970 was a functorial and geometrically meaningful definition of K_n for all $n \geq 3$, together with long exact sequences and comparison theorems robust enough to be used in algebraic geometry. Quillen's breakthrough was to replace the search for individual groups by the construction of a space whose homotopy groups are the higher K-groups. For exact categories this space is produced from a category $Q\mathcal{M}$ built out of admissible subquotients, and for rings it is identified with the classical $B\mathrm{GL}(R)^+$ space.

The remarkable point is that Quillen's theory is not merely a definition. It comes equipped with a homotopy-theoretic fiber formalism and with structural theorems—resolution, dévissage, and localization—which promote K-theory from an isolated invariant into a machine.

2. Exact categories as the natural input

Quillen works with *exact categories*, not just abelian categories. This is the correct level of generality if one wants to treat projective modules, vector bundles, and extension-closed subcategories of abelian categories on equal footing.

Definition 2.1. An *exact category* is an additive category $\mathcal{M}$ endowed with a class of kernel–cokernel pairs

$$0 \longrightarrow M' \xrightarrow{i} M \xrightarrow{j} M'' \longrightarrow 0$$

called *admissible exact sequences*, satisfying the closure properties needed for pullback and pushout along admissible monomorphisms and epimorphisms. Morphisms i and j in such a sequence are called *admissible monomorphisms* and *admissible epimorphisms*, respectively.

Abelian categories are exact in the evident way. Any additive category is also exact in the split sense by declaring split short exact sequences to be admissible. Thus the category $P(R)$ of finitely generated projective R-modules becomes an exact category although it is usually not abelian.

A first conceptual point is that exactness here is not an afterthought. Quillen's higher theory only depends on the exact structure, not on a chosen ambient abelian category. That feature is already visible in *Higher algebraic K-theory: I*.

3. Classifying spaces of categories

For a small category $\mathcal{C}$, write $N\mathcal{C}$ for its nerve and $\mathrm{B}\mathcal{C} := |N\mathcal{C}|$ for the geometric realization. Quillen's philosophy is that the correct higher invariants are the homotopy groups of such classifying spaces.

Two elementary examples guide everything:

1. If J is a poset, then $\mathrm{B}J$ is the geometric realization of its order complex.
2. If G is a group viewed as a one-object category, then $\mathrm{B}G$ is the usual Eilenberg–MacLane space $K(G,1)$.

So classifying spaces interpolate between combinatorics and homotopy theory. Quillen exploits this in a systematically categorical way.

3.1. Quillen's homotopy-theoretic input: Theorem B.

The theorem that makes the machine work is Quillen's Theorem B. In modern language it identifies homotopy fibers of maps of classifying spaces induced by functors, under a base-change hypothesis.

Let $f\colon \mathcal{C} \to \mathcal{C}'$ be a functor and $Y \in \mathcal{C}'$. Denote by $Y\backslash f$ the comma category of arrows $Y \to f(X)$.

Theorem 3.1 (Quillen's Theorem B, informal form)**.** *Assume that for every arrow $u\colon Y \to Y'$ in $\mathcal{C}'$, the induced functor*

$$Y'\backslash f \longrightarrow Y\backslash f$$

is a homotopy equivalence. Then for each object Y the classifying space $\mathrm{B}(Y\backslash f)$ models the homotopy fiber of

$$\mathrm{B}f\colon \mathrm{B}\mathcal{C} \longrightarrow \mathrm{B}\mathcal{C}'$$

over the point $Y \in \mathrm{B}\mathcal{C}'$.

This theorem yields long exact homotopy sequences and is the basic reason exact functors produce fiber sequences in K-theory. In Quillen's paper it is the hidden engine behind localization.

$$\begin{tikzcd} Y\backslash f \arrow[r] \arrow[d] & \mathcal{C} \arrow[d, "f"] \\ * \arrow[r, "Y"] & \mathcal{C}' \end{tikzcd}$$

When the square is homotopy-cartesian, $\mathrm{B}(Y\backslash f)$ identifies the homotopy fiber of Bf at Y.

4. The Q-construction

We now reach the heart of higher Quillen theory.

4.1. **Definition of $Q\mathcal{M}$.** Let $\mathcal{M}$ be a small exact category. The category $Q\mathcal{M}$ has the same objects as $\mathcal{M}$. A morphism from M to M' is represented by a diagram

$$M \xleftarrow{j} N \xrightarrow{i} M'$$

where j is an admissible epimorphism and i is an admissible monomorphism, modulo isomorphism of the middle object inducing the identity on source and target.

Intuitively, such a morphism identifies M with an admissible subquotient of M'. Indeed, if we write the admissible layer in M' determined by i as

$$M'_0 \subseteq M'_1 \subseteq M',$$

then the subquotient M'_1/M'_0 is isomorphic to M.

Composition is defined by pullback. If

$$M \xleftarrow{j} N \xrightarrow{i} M', \qquad M' \xleftarrow{j'} N' \xrightarrow{i'} M''$$

represent two morphisms, their composite is represented by the pullback square

$$\begin{tikzcd} P \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & N' \arrow[d, "j'"'] \\ N \arrow[r, "i"] & M' \end{tikzcd}$$

and by the span

$$M \xleftarrow{j\circ\mathrm{pr}_1} P \xrightarrow{i'\circ\mathrm{pr}_2} M''.$$

This is well-defined because admissible epimorphisms are stable under pullback along admissible monomorphisms in an exact category.

4.2. **Injective and surjective morphisms in $Q\mathcal{M}$.** Every admissible monomorphism $i\colon M' \hookrightarrow M$ gives a morphism

$$i^!\colon M' \to M$$

in $Q\mathcal{M}$, and every admissible epimorphism $j\colon M \twoheadrightarrow M''$ gives a morphism

$$j_!\colon M'' \to M.$$

Every morphism in $Q\mathcal{M}$ factors, uniquely up to unique isomorphism, as an injective morphism followed by a surjective one. This factorization reflects the idea that every subquotient may be built in two steps: first pass to a quotient, then embed the result.

$$\begin{tikzcd} N' \arrow[r, "i'"] \arrow[d, "j'"'] & M'' \\ N \arrow[r, "i"'] & M' \end{tikzcd} \qquad u = j'_!\, i^!.$$

This rigid factorization is one of the reasons $Q\mathcal{M}$ is rich enough to encode exact sequences, yet still combinatorial enough to have a workable classifying space.

5. The definition of higher K-groups

Fix a zero object $0 \in \mathcal{M}$. Quillen defines the higher groups by

$$K_i(\mathcal{M}) := \pi_{i+1}(BQ\mathcal{M}, 0), \qquad i \geq 0.$$

The shift by one is deliberate: the fundamental group of $BQ\mathcal{M}$ recovers the classical Grothendieck group.

Theorem 5.1 (Quillen)**.** *There is a canonical isomorphism*

$$\pi_1(BQ\mathcal{M}, 0) \cong K_0(\mathcal{M}),$$

where $K_0(\mathcal{M})$ is the Grothendieck group generated by isomorphism classes $[M]$ modulo the relations

$$[M] = [M'] + [M'']$$

for each admissible exact sequence

$$0 \to M' \to M \to M'' \to 0.$$

Thus Quillen's definition really extends the classical theory. The point is not only that K_0 appears in degree zero, but that the entire tower of groups is now governed by a single delooping space.

Remark 5.2. For a ring R, one applies this to the exact category $P(R)$ of finitely generated projective modules and writes

$$K_i(R) := K_i(P(R)).$$

For a noetherian ring one similarly defines $K'_i(R)$ by using the abelian category of finitely generated modules. For regular rings the resolution theorem identifies these two theories.

6. Additivity and characteristic exact sequences

A first nontrivial structural fact is that exact sequences of exact functors induce additivity relations on K-theory. At the level of K_0 this is obvious; Quillen proves that the same phenomenon persists in all degrees.

Suppose

$$0 \to F' \to F \to F'' \to 0$$

is an exact sequence of exact functors between exact categories. Then the induced maps on K_i satisfy

$$F_* = F'_* + F''_*.$$

More generally, an exact complex of exact functors gives an alternating sum relation on higher K-groups. This is the higher additivity theorem hidden in Quillen's Section 3.

One conceptual reading is the following: the classifying space functor does not preserve exact sequences on the nose, but after passing to homotopy groups the exact structure becomes additive.

7. Resolution theorem

The resolution theorem explains when a full exact subcategory computes the same K-theory as the ambient one.

Theorem 7.1 (Resolution theorem, schematic form)**.** *Let $\mathcal{P} \subseteq \mathcal{M}$ be a full exact subcategory, closed under extensions, such that:*

(i) whenever

$$0 \to M' \to M \to M'' \to 0$$

is exact in $\mathcal{M}$ and $M, M'' \in \mathcal{P}$, then $M' \in \mathcal{P}$;

(ii) every object $M \in \mathcal{M}$ admits an admissible epimorphism $P \twoheadrightarrow M$ with $P \in \mathcal{P}$.

Then the inclusion $\mathcal{P} \hookrightarrow \mathcal{M}$ induces a homotopy equivalence

$$BQ\mathcal{P} \simeq BQ\mathcal{M},$$

hence isomorphisms $K_i(\mathcal{P}) \cong K_i(\mathcal{M})$ for all $i \geq 0$.

This theorem is the algebraic-geometric bridge between vector bundles and coherent sheaves on regular schemes. If every coherent sheaf admits a finite resolution by vector bundles, then K-theory and K'-theory agree.

$$\begin{tikzcd} \mathcal{P} \arrow[r, hook] \arrow[d] & \mathcal{M} \arrow[d] \\ BQ\mathcal{P} \arrow[r] & BQ\mathcal{M} \end{tikzcd}$$

In modern language: exact resolutions produce equivalences of K-theory spaces.

8. Dévissage

Dévissage is one of Quillen's deepest ideas. It says that when every object has a finite filtration with subquotients in a smaller abelian subcategory, the smaller category already captures the whole K-theory.

Theorem 8.1 (Dévissage)**.** *Let $\mathcal{B}$ be a non-empty full subcategory of an abelian category $\mathcal{A}$, closed under subobjects, quotient objects, and finite products. Assume that every object $M \in \mathcal{A}$ admits a finite filtration*

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M$$

with each quotient M_j/M_{j-1} in $\mathcal{B}$. Then the inclusion induces a homotopy equivalence

$$BQ\mathcal{B} \xrightarrow{\simeq} BQ\mathcal{A},$$

so that

$$K_i(\mathcal{B}) \cong K_i(\mathcal{A}) \qquad \text{for all } i \geq 0.$$

The standard consequence is that if $\mathcal{A}$ is a finite length abelian category with simple objects $\{X_j\}_{j\in J}$ and division endomorphism rings $D_j = \mathrm{End}(X_j)^{\mathrm{op}}$, then

$$K_i(\mathcal{A}) \cong \bigoplus_{j\in J} K_i(D_j).$$

For Artinian and noetherian categories this turns structural composition series directly into K-theory calculations.

9. Localization

Localization is the theorem that turns higher K-theory into a genuine cohomological machine.

Let $\mathcal{B}$ be a Serre subcategory of an abelian category $\mathcal{A}$, and let $\mathcal{A}/\mathcal{B}$ be the Gabriel quotient. Quillen proves that there is a long exact sequence

$$\cdots \to K_i(\mathcal{B}) \to K_i(\mathcal{A}) \to K_i(\mathcal{A}/\mathcal{B}) \xrightarrow{\partial} K_{i-1}(\mathcal{B}) \to \cdots \to K_0(\mathcal{A}/\mathcal{B}).$$

Theorem 9.1 (Localization)**.** *If $\mathcal{B}$ is a Serre subcategory of the abelian category $\mathcal{A}$, then the sequence above is exact and functorial.*

The homotopy-theoretic content is that the map

$$BQ\mathcal{B} \longrightarrow BQ\mathcal{A}$$

fits into a homotopy fiber sequence after suitable identification of the fiber with the quotient category. This is precisely where Quillen's Theorem B enters decisively.

$$BQ\mathcal{B} \longrightarrow BQ\mathcal{A} \longrightarrow BQ(\mathcal{A}/\mathcal{B})$$

and therefore

$$\cdots \to \pi_{n+1} BQ(\mathcal{A}/\mathcal{B}) \to \pi_n BQ\mathcal{B} \to \pi_n BQ\mathcal{A} \to \cdots$$

becomes the long exact localization sequence in K-theory.

10. The $B\mathrm{GL}^+$ construction for rings

For a ring R, Quillen's original topological definition is based on the infinite general linear group

$$\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R)$$

and its elementary subgroup $E(R) \subseteq \mathrm{GL}(R)$, which is perfect. The plus-construction yields a space

$$B\mathrm{GL}(R)^+$$

with the following properties:

(1) $\pi_1(B\mathrm{GL}(R)^+) \cong \mathrm{GL}(R)/E(R) = K_1(R)$;
(2) the map $B\mathrm{GL}(R) \to B\mathrm{GL}(R)^+$ induces isomorphisms on homology with local coefficients for $K_1(R)$-modules.

Then one defines

$$K_n(R) := \pi_n\big(K_0(R) \times B\mathrm{GL}(R)^+\big), \qquad n \geq 0.$$

For $n \geq 1$ this is just $\pi_n(B\mathrm{GL}(R)^+)$. By construction this recovers classical K_1, and Quillen showed that it also recovers classical K_2.

11. The comparison theorem $+ = Q$

The most elegant bridge in the entire subject is the comparison between the Q-construction and the plus-construction. For split exact categories the Q-construction agrees with the group-completion construction. Applied to $P(R)$, this identifies Quillen's two ways of building higher K-theory.

Theorem 11.1 (Quillen, $+ = Q$)**.** *For every ring R there is a homotopy equivalence*

$$\Omega BQP(R) \simeq K_0(R) \times B\mathrm{GL}(R)^+.$$

Consequently,

$$K_n(P(R)) \cong K_n(R) \qquad \textit{for all } n \geq 0.$$

11.1. **Mechanism of the proof.** The proof proceeds through an auxiliary category $E\mathcal{A}$ of short exact sequences in the split exact category $\mathcal{A}$. There is a target functor

$$t\colon E\mathcal{A} \to Q\mathcal{A}, \qquad (A \hookrightarrow B \twoheadrightarrow C) \longmapsto C.$$

For each object $C \in \mathcal{A}$, the fiber category $E_C = t^{-1}(C)$ carries a symmetric monoidal structure. When $\mathcal{A}$ is split exact, each E_C is equivalent enough to the groupoid $S = \mathrm{iso}(\mathcal{A})$ that one obtains homotopy equivalences

$$S^{-1}S \xrightarrow{\simeq} S^{-1}E_C.$$

Quillen's Theorem B then shows that

$$S^{-1}S \longrightarrow S^{-1}E\mathcal{A} \longrightarrow Q\mathcal{A}$$

is a homotopy fibration. Since $E\mathcal{A}$ is contractible, so is $S^{-1}E\mathcal{A}$, and therefore

$$\Omega BQ\mathcal{A} \simeq B(S^{-1}S).$$

For $\mathcal{A} = P(R)$, the group completion theorem identifies $B(S^{-1}S)$ with $K_0(R) \times B\mathrm{GL}(R)^+$, producing the desired equivalence.

$$S^{-1}S \longrightarrow S^{-1}E\mathcal{A} \xrightarrow{T} Q\mathcal{A}$$

with the middle term contractible in the split exact case. Looping the base therefore recovers the fiber.

12. Why Quillen's higher theory was decisive

At this point one can state more sharply what Quillen changed. Before higher K-theory, one had a very successful low-dimensional collection of constructions, but one did not yet have a *homotopy-theoretic formalism* capable of transporting exactness phenomena from algebra and geometry into long exact sequences of invariants. Quillen's contribution was decisive for at least four mathematical reasons.

12.1. **A uniform homotopy-theoretic definition.** The definition

$$K_i(\mathcal{M}) = \pi_{i+1}(BQ\mathcal{M}, 0), \qquad i \geq 0,$$

replaces a list of unrelated recipes by one delooped object. In particular:

(a) K_0 is recovered as $\pi_1(BQ\mathcal{M})$;
(b) exact functors induce continuous maps of classifying spaces and hence morphisms on all higher groups;
(c) comparison statements become statements about equivalences or fiber sequences of spaces;
(d) exactness in the category-theoretic input becomes homotopy exactness after applying BQ.

This point is not merely philosophical. It is the reason why one can deduce localization sequences, spectral sequences, transfer maps, and comparison theorems from a single framework rather than inventing them independently in each degree.

12.2. **The passage from exact categories to homotopy fibers.** The structural theorem behind this is Quillen's Theorem B. Its role in the theory can be summarized by the following principle.

> Whenever an exact functor admits a good categorical description of its fibers, Theorem B upgrades that description to a homotopy fiber sequence of classifying spaces, hence to a long exact sequence in K-groups.

That principle is what turns the inclusion of a Serre subcategory into the localization sequence, and later turns the filtration by support into a spectral sequence. Thus the genuinely new object is not just K_n, but the possibility of doing homotopy theory directly with exact categories.

12.3. **Why the comparison $+ = Q$ matters conceptually.** The theorem

$$\Omega BQP(R) \simeq K_0(R) \times B\mathrm{GL}(R)^+$$

shows that Quillen's exact-categorical and topological constructions are not competing definitions but two manifestations of the same higher invariant. From the left-hand side one sees functoriality for exact categories, filtrations, and quotients; from the right-hand side one sees the stable linear group and the classical Whitehead-theoretic origin of K_1 and K_2. The comparison therefore explains simultaneously:

(1) why the new theory extends Bass–Milnor low-dimensional K-theory;
(2) why it is intrinsically homotopy-theoretic;
(3) why the stable linear group governs the split exact case.

12.4. **Why the theory became geometric immediately.** Because localization and dévissage are built into the theory, one can filter coherent sheaves by codimension of support and obtain a spectral sequence whose E_1-page is expressed in terms of residue fields. This is the point where the subject ceases to be merely an algebraic theory of projective modules and becomes a geometric theory sensitive to the stratification of a scheme by codimension. In Quillen's foundational paper this already leads to the support spectral sequence, the equicharacteristic cases of Gersten's conjecture, and Bloch's formula for Chow groups modulo rational equivalence. These are not peripheral applications; they are evidence that the higher theory was the correct ambient object from the start.

13. Schemes and the geometric payoff

We now expand the geometric part in greater detail. This is the place where one sees most clearly why Quillen's higher theory is powerful: exact categories and localization produce a filtration of coherent sheaves by support, and from that filtration one obtains a spectral sequence, a Gersten complex, and finally Bloch's formula.

13.1. **K and K' of a scheme.** Let X be a noetherian scheme. Write $\mathrm{Vect}(X)$ for the exact category of vector bundles on X and $\mathrm{Coh}(X)$ for the abelian category of coherent sheaves. Quillen defines

$$K_i(X) := K_i(\mathrm{Vect}(X)), \qquad K'_i(X) := K_i(\mathrm{Coh}(X)).$$

There is a natural map

$$K_i(X) \longrightarrow K'_i(X)$$

induced by the inclusion of vector bundles into coherent sheaves. If X is regular, every coherent sheaf admits a finite resolution by vector bundles, so by the resolution theorem this map is an isomorphism.

Theorem 13.1 (Resolution for regular schemes)**.** *If X is regular and noetherian, then for every $i \geq 0$ the canonical map*

$$K_i(X) \xrightarrow{\sim} K'_i(X)$$

is an isomorphism.

Proof. We verify the hypotheses of the resolution theorem for the inclusion

$$\mathrm{Vect}(X) \hookrightarrow \mathrm{Coh}(X).$$

First, $\mathrm{Vect}(X)$ is extension-closed inside $\mathrm{Coh}(X)$. Indeed, let

$$0 \longrightarrow \mathcal{E}' \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}'' \longrightarrow 0$$

be an exact sequence of coherent sheaves with $\mathcal{E}$ and $\mathcal{E}''$ locally free. Fix a point $x \in X$ and localize at x. Since $\mathcal{E}_x$ and $\mathcal{E}''_x$ are free modules over the local ring $\mathcal{O}_{X,x}$ and $\mathcal{E}''_x$ is projective, the sequence splits after localization. Hence $\mathcal{E}'_x$ is a direct summand of a free module, therefore projective, hence free because $\mathcal{O}_{X,x}$ is local. Thus $\mathcal{E}'$ is locally free.

Second, every coherent sheaf on a regular noetherian scheme admits a finite locally free resolution. Concretely, if $\mathcal{F} \in \mathrm{Coh}(X)$ and $x \in X$, the local ring $\mathcal{O}_{X,x}$ is regular; therefore every finitely generated $\mathcal{O}_{X,x}$-module has finite projective dimension. Choosing a surjection from a finite rank vector bundle onto $\mathcal{F}$, taking kernels, and iterating, one obtains locally a finite resolution by free modules. Noetherianity allows these local constructions to be patched into a finite globally defined locally free resolution

$$0 \longrightarrow \mathcal{E}_m \longrightarrow \cdots \longrightarrow \mathcal{E}_0 \longrightarrow \mathcal{F} \longrightarrow 0.$$

The two conditions above are exactly the hypotheses of Quillen's resolution theorem. Therefore the inclusion $\mathrm{Vect}(X) \hookrightarrow \mathrm{Coh}(X)$ induces a homotopy equivalence

$$BQ\mathrm{Vect}(X) \xrightarrow{\simeq} BQ\mathrm{Coh}(X).$$

Passing to homotopy groups yields canonical isomorphisms

$$K_i(X) = K_i(\mathrm{Vect}(X)) \cong K_i(\mathrm{Coh}(X)) = K'_i(X)$$

for every $i \geq 0$, as claimed. □

Thus for regular schemes one may pass freely between vector bundles and coherent sheaves according to convenience. The K'-theory language is however better adapted to support conditions and hence to the coniveau filtration.

13.2. **Closed immersions and localization on schemes.** Let $i\colon Z \hookrightarrow X$ be a closed immersion with open complement $j\colon U \hookrightarrow X$. Denote by $\mathrm{Coh}_Z(X)$ the Serre subcategory of coherent sheaves on X supported on Z. Gabriel's theorem identifies the quotient category $\mathrm{Coh}(X)/\mathrm{Coh}_Z(X)$ with $\mathrm{Coh}(U)$. Quillen's localization theorem therefore yields a long exact sequence

$$\cdots \to K'_i(Z) \xrightarrow{i_*} K'_i(X) \xrightarrow{j^*} K'_i(U) \xrightarrow{\partial} K'_{i-1}(Z) \to \cdots .$$

Theorem 13.2 (Localization for a closed subscheme)**.** *Let X be noetherian and let $Z \subseteq X$ be a closed subscheme with open complement U. Then there is a natural long exact sequence*

$$\cdots \to K'_i(Z) \to K'_i(X) \to K'_i(U) \to K'_{i-1}(Z) \to \cdots \to K'_0(U).$$

If X is regular and Z is also regular, the same statement may be written in terms of K_i.

Proof. The category $\mathrm{Coh}_Z(X)$ is a Serre subcategory of $\mathrm{Coh}(X)$: if

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is exact in $\mathrm{Coh}(X)$ and $\mathcal{F}$ is supported on Z, then the supports of $\mathcal{F}'$ and $\mathcal{F}''$ are contained in the support of $\mathcal{F}$, hence are also contained in Z. Thus $\mathrm{Coh}_Z(X)$ is stable under subobjects, quotients, and extensions.

Next, restriction to the open subscheme $U = X \setminus Z$ induces an exact functor

$$j^*\colon \mathrm{Coh}(X) \longrightarrow \mathrm{Coh}(U).$$

Its kernel is exactly $\mathrm{Coh}_Z(X)$, because a coherent sheaf restricts to zero on U if and only if its support is contained in Z. Gabriel's theorem identifies the Serre quotient with the category of coherent sheaves on the open complement:

$$\mathrm{Coh}(X)/\mathrm{Coh}_Z(X) \simeq \mathrm{Coh}(U).$$

Therefore Quillen's localization theorem applies to the inclusion of exact categories

$$\mathrm{Coh}_Z(X) \hookrightarrow \mathrm{Coh}(X).$$

It provides a homotopy fiber sequence

$$BQ\mathrm{Coh}_Z(X) \longrightarrow BQ\mathrm{Coh}(X) \longrightarrow BQ\big(\mathrm{Coh}(X)/\mathrm{Coh}_Z(X)\big) \simeq BQ\mathrm{Coh}(U).$$

Taking homotopy groups and using the convention $K'_i(X) = \pi_{i+1}(BQ\mathrm{Coh}(X))$ yields the long exact sequence

$$\cdots \to K'_i(Z) \to K'_i(X) \to K'_i(U) \to K'_{i-1}(Z) \to \cdots .$$

When X and Z are regular, the resolution theorem identifies K and K' on both schemes, so the same exact sequence may be written with K_i in place of K'_i. □

The scheme-theoretic localization sequence is therefore literally the categorical localization theorem applied to coherent sheaves.

13.3. **Filtration by codimension of support.** For each integer $p \geq 0$, let $\mathcal{M}^p(X) \subseteq \mathrm{Coh}(X)$ denote the full Serre subcategory of coherent sheaves whose support has codimension at least p. Thus

$$\mathrm{Coh}(X) = \mathcal{M}^0(X) \supseteq \mathcal{M}^1(X) \supseteq \mathcal{M}^2(X) \supseteq \cdots .$$

This descending filtration is the geometric incarnation of the coniveau filtration.

The successive quotients of this filtration are described by dévissage. Intuitively, when one passes from codimension $\geq p$ to codimension $\geq p+1$, only the strata of exact codimension p survive, and near a point $x \in X^{(p)}$ the relevant abelian category becomes the category of finite-length modules over the local ring $\mathcal{O}_{X,x}$, whose K-theory is the same as that of the residue field $k(x)$.

Proposition 13.3 (Successive quotients of the support filtration)**.** *Let X be noetherian. Then the quotient exact category $\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)$ is equivalent, after passage to K-theory, to the coproduct of the finite-length module categories at codimension-p points. Consequently,*

$$K_i\big(\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)\big) \cong \bigoplus_{x \in X^{(p)}} K_i(k(x)).$$

Proof. Fix $p \geq 0$. By definition, an object of the quotient category

$$\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)$$

is represented by a coherent sheaf whose support has codimension at least p, two such sheaves being identified if their difference is supported in codimension at least $p+1$. Thus only the codimension-p part of the support is visible in the quotient.

Let $\mathcal{F} \in \mathcal{M}^p(X)$. The set of generic points of irreducible components of $\mathrm{Supp}(\mathcal{F})$ having codimension exactly p is finite; denote it by

$$\{x_1, \ldots, x_m\} \subseteq X^{(p)}.$$

There is a standard finite filtration of $\mathcal{F}$ in $\mathrm{Coh}(X)$ whose successive quotients are supported either in codimension $\geq p+1$ or on the closures $\overline{\{x_i\}}$. After passage to the quotient by $\mathcal{M}^{p+1}(X)$, only the pieces corresponding to the points x_i remain. This shows that the quotient category decomposes, up to the level relevant for K-theory, into the direct sum of its local contributions at codimension-p points.

For a fixed $x \in X^{(p)}$, let $\mathcal{C}_x$ be the full subcategory of $\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)$ generated by objects whose support has generic point x. Localizing at x identifies $\mathcal{C}_x$ with the exact category of finite-length $\mathcal{O}_{X,x}$-modules. Indeed, once all components of larger codimension have been killed, the only remaining information is the finite-length module carried by the stalk at x. Hence

$$K_i(\mathcal{C}_x) \cong K_i\big(\mathrm{fl}(\mathcal{O}_{X,x})\big),$$

where $\mathrm{fl}(\mathcal{O}_{X,x})$ denotes the abelian category of finite-length modules.

Now apply Quillen's dévissage theorem to the abelian category $\mathrm{fl}(\mathcal{O}_{X,x})$. Its simple objects are precisely the finite-dimensional $k(x)$-vector spaces, and every finite-length module admits a finite filtration whose subquotients are copies of the residue field $k(x)$. Therefore the inclusion

$$\mathrm{Vect}_f(k(x)) \hookrightarrow \mathrm{fl}(\mathcal{O}_{X,x})$$

induces an equivalence on K-theory. In particular,

$$K_i\big(\mathrm{fl}(\mathcal{O}_{X,x})\big) \cong K_i(k(x)).$$

Since the codimension-p points contribute independently, the quotient category yields a direct sum decomposition on K-groups:

$$K_i\big(\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)\big) \cong \bigoplus_{x\in X^{(p)}} K_i\big(\mathrm{fl}(\mathcal{O}_{X,x})\big) \cong \bigoplus_{x\in X^{(p)}} K_i(k(x)).$$

This is the desired identification. □

This identification is the decisive input for the spectral sequence.

13.4. **The coniveau spectral sequence.** Apply localization to the inclusions

$$\mathcal{M}^{p+1}(X) \hookrightarrow \mathcal{M}^p(X).$$

For each p this yields an exact couple, and the resulting exact couple of the filtration gives a spectral sequence.

Theorem 13.4 (Quillen's support spectral sequence)**.** *Let X be a noetherian scheme and let $X^{(p)}$ denote the set of points of codimension p. Then there is a spectral sequence*

$$E_1^{p,q}(X) = \bigoplus_{x\in X^{(p)}} K_{-p-q}(k(x)) \Longrightarrow K'_{-p-q}(X).$$

If X has finite Krull dimension, the spectral sequence is strongly convergent. It is contravariant for flat morphisms.

Construction. For each p one has an exact sequence of exact categories

$$\mathcal{M}^{p+1}(X) \longrightarrow \mathcal{M}^p(X) \longrightarrow \mathcal{M}^p(X)/\mathcal{M}^{p+1}(X).$$

Applying K-theory gives long exact sequences

$$\cdots \to K_n\big(\mathcal{M}^{p+1}(X)\big) \to K_n\big(\mathcal{M}^p(X)\big) \to K_n\big(\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)\big) \to K_{n-1}\big(\mathcal{M}^{p+1}(X)\big) \to \cdots.$$

These long exact sequences assemble into an exact couple. Using the previous proposition to identify the quotient terms yields

$$D_1^{p,q} = K_{-p-q}\big(\mathcal{M}^p(X)\big), \qquad E_1^{p,q} = \bigoplus_{x\in X^{(p)}} K_{-p-q}(k(x)).$$

The spectral sequence associated with this exact couple abuts to $K'_{-p-q}(X) = K_{-p-q}(\mathcal{M}^0(X))$. □

The associated exact couple may be visualized as follows:

$$\begin{tikzcd}
D_1^{p+1,q-1} \arrow[rr, "i"] & & D_1^{p,q} \arrow[dl, "j"] \\
& E_1^{p,q} \arrow[ul, "k"] &
\end{tikzcd}$$

where i is induced by inclusion, j by passage to the quotient, and k is the connecting homomorphism. The first differential is the composite $d_1 = j \circ k$.

13.5. **The Gersten problem.** The support spectral sequence suggests a natural exact complex built from residue fields. The question is whether, for a regular local ring, the E_1-page already resolves the K-theory sheaf. Quillen formulates this as a condition on the support filtration.

Proposition 13.5 (Equivalent forms of the Gersten condition)**.** *Let X be noetherian. The following are equivalent.*

(i) *For every $p \geq 0$, the inclusion*
$$\mathcal{M}^{p+1}(X) \hookrightarrow \mathcal{M}^p(X)$$
induces the zero map on all K-groups.

(ii) *For every $q \geq 0$, the edge homomorphism from the support spectral sequence identifies $K_q(X)$ with the E_2-term in filtration degree 0.*

(iii) *The Gersten complex obtained from the differential d_1 on the E_1-page is exact:*
$$0 \to \bigoplus_{x \in X^{(0)}} K_n(k(x)) \xrightarrow{d_1} \bigoplus_{x \in X^{(1)}} K_{n-1}(k(x)) \xrightarrow{d_1} \cdots \xrightarrow{d_1} \bigoplus_{x \in X^{(n)}} K_0(k(x)).$$

Proof. Recall the exact couple attached to the support filtration:
$$D_1^{p,q} = K_{-p-q}(\mathcal{M}^p(X)), \qquad E_1^{p,q} = K_{-p-q}\big(\mathcal{M}^p(X)/\mathcal{M}^{p+1}(X)\big),$$
with structure maps
$$i\colon D_1^{p+1,q-1} \to D_1^{p,q}, \qquad j\colon D_1^{p,q} \to E_1^{p,q}, \qquad k\colon E_1^{p,q} \to D_1^{p+1,q-1}.$$
The map i is induced by the inclusion $\mathcal{M}^{p+1}(X) \hookrightarrow \mathcal{M}^p(X)$, and the first differential is $d_1 = j \circ k$.

Assume first (i). Then every map i in the exact couple is zero already on the D_1-terms. When one derives the exact couple, all higher i_r are therefore zero as well. Consequently there are no differentials beyond d_1, and
$$E_2^{p,q} = H(E_1^{\bullet,q}, d_1)$$
is the final value of the spectral sequence. In particular, the spectral sequence degenerates at E_2. Since the abutment is $K'_{-p-q}(X)$ and all filtration maps vanish, the only possible nonzero limit terms lie in filtration degree $p = 0$. Therefore $E_2^{p,q} = 0$ for $p \neq 0$, and the edge map
$$K'_{-q}(X) \longrightarrow E_2^{0,q}(X)$$
is an isomorphism. This proves (ii).

Now compare (ii) and (iii). By construction of the support spectral sequence and by the identification of the E_1-page with residue-field K-groups, the row computing $E_2^{p,-n-p}$ is precisely the Gersten complex
$$0 \to K'_n(X) \xrightarrow{e} \bigoplus_{x \in X^{(0)}} K_n(k(x)) \xrightarrow{d_1} \bigoplus_{x \in X^{(1)}} K_{n-1}(k(x)) \xrightarrow{d_1} \cdots .$$
Thus $E_2^{p,-n-p}$ is the p-th homology of that complex. Saying that the edge map is an isomorphism and that all $E_2^{p,q}$ with $p \neq 0$ vanish is exactly the statement that the Gersten complex is exact. Hence (ii) and (iii) are equivalent.

Finally assume (ii). Since the spectral sequence is concentrated in filtration degree 0 from the E_2-page on, the filtration on the abutment is trivial and the maps from higher filtration pieces to lower ones must vanish. Concretely, the image of
$$K_*(\mathcal{M}^{p+1}(X)) \longrightarrow K_*(\mathcal{M}^p(X))$$

controls the next stage of the filtration on the abutment; if some such image were nonzero, a nontrivial class would survive in positive filtration degree, contradicting (ii). Therefore the maps induced by the inclusions $\mathcal{M}^{p+1}(X) \hookrightarrow \mathcal{M}^p(X)$ are zero, proving (i).

Thus (i), (ii), and (iii) are equivalent. □

For a regular local ring one expects the support filtration to be as transverse as possible, so that the Gersten complex is exact. This is the content of the Gersten conjecture in Quillen's paper.

> **Gersten conjecture.** If A is a regular local ring, then the support filtration on $\mathrm{Spec}A$ satisfies the equivalent conditions above.

13.6. **Quillen's equicharacteristic proof: strategy.** We now explain in detail the proof in the equicharacteristic cases established by Quillen. The result proved in the paper is the following.

Theorem 13.6 (Quillen, equicharacteristic semi-local case)**.** *Let R be a finite type algebra over a field k, let S be a finite set of prime ideals such that $R_{\mathfrak{p}}$ is regular for each $\mathfrak{p} \in S$, and let $A = S^{-1}R$ be the resulting regular semi-local ring. Then* $\mathrm{Spec}A$ *satisfies the Gersten condition.*

Quillen also proves the analogous statement for formal power series rings $k[[x_1, \dots, x_n]]$ and for convergent power series rings over complete valued fields.

The proof has a beautiful shape. It is not a brute-force verification of exactness of the Gersten complex. Instead, one proves that the filtration maps

$$K_*\big(\mathcal{M}^{p+1}(A)\big) \longrightarrow K_*\big(\mathcal{M}^p(A)\big)$$

vanish for all p. Once this is known, the Gersten complex and its exactness follow formally from the spectral sequence.

13.7. **Step 1: reduction to the smooth finite type case.** We begin with the statement of Theorem 5.11 in Quillen's notation. Let R be a finite type algebra over a field k, let $S \subseteq \mathrm{Spec}R$ be finite, assume that each local ring $R_{\mathfrak{p}}$ for $\mathfrak{p} \in S$ is regular, and put $A = S^{-1}R$. The goal is to show that for every $p \geq 0$ the inclusion

$$\mathcal{M}^{p+1}(A) \hookrightarrow \mathcal{M}^p(A)$$

induces the zero map on all K-groups.

The first reduction replaces the possibly huge field k by a finitely generated subfield. Choose a subfield $k' \subseteq k$, finitely generated over the prime field, together with a finite type k'-algebra R' and a finite subset $S' \subseteq \mathrm{Spec}R'$ such that

$$R \cong k \otimes_{k'} R'$$

and the primes in S are obtained from the primes in S' by base change. If A' denotes the localization of R' at S', then

$$A \cong k \otimes_{k'} A'.$$

Let k_i run through the finitely generated subfields of k containing k'. The ring A is the filtered colimit of the rings $k_i \otimes_{k'} A'$, and Quillen's limit statement for the support filtration implies

$$K_*\big(\mathcal{M}^p(A)\big) \cong \varinjlim_i K_*\big(\mathcal{M}^p(k_i \otimes_{k'} A')\big).$$

Therefore it is enough to prove the theorem after replacing k by a finitely generated extension of the prime field. Enlarging R' if necessary, one may then assume that k itself is the prime field.

This reduction is useful because prime fields are perfect. Since $A = S^{-1}R$ is regular, the ring R is regular at each point of S. Over a perfect field, regularity and smoothness are equivalent for finite type algebras at a given point; hence R is smooth over k at each point of S. Smoothness is an open condition, so there exists $f \in R$ not vanishing at any point of S such that R_f is smooth over k. Replacing R by R_f does not change the semi-local ring $A = S^{-1}R$, because all primes in S survive the localization. Thus we may and do assume from the outset that

$$R \text{ is a smooth finite type algebra over a perfect field } k, \text{ and } A = S^{-1}R.$$

After this reduction, the whole problem becomes geometric: one must prove that the support filtration on a smooth affine scheme becomes invisible on K-groups after passing to the semi-local ring determined by S.

13.8. **Step 2: reduction to a regular hypersurface section.** The second reduction explains why it is enough to study coherent sheaves living on principal hypersurfaces. Since

$$A = S^{-1}R = \varinjlim_f R_f,$$

where f runs through elements of R not vanishing at any point of S, Quillen's compatibility of the support filtration with filtered colimits gives

$$K_*\big(\mathcal{M}^{p+1}(A)\big) \cong \varinjlim_f K_*\big(\mathcal{M}^{p+1}(R_f)\big).$$

Consequently, to prove that

$$K_*\big(\mathcal{M}^{p+1}(A)\big) \longrightarrow K_*\big(\mathcal{M}^{p}(A)\big)$$

vanishes, it is enough to show that for each class in some $K_*\big(\mathcal{M}^{p+1}(R_f)\big)$ one can localize further so that its image in $K_*\big(\mathcal{M}^{p}(R_{ff'})\big)$ is zero.

The filtration by support allows one to write $\mathcal{M}^{p+1}(R)$ as a filtered union of subcategories coming from closed subsets of codimension at least $p+1$. On a regular ring, every codimension-$(p+1)$ generic point is cut out locally by a regular element. Said differently, after shrinking around the finite set S, an object of $\mathcal{M}^{p+1}$ may be viewed as a codimension-p object on a hypersurface. Quillen packages this observation in the identity

$$K_*\big(\mathcal{M}^{p+1}(R)\big) \cong \varinjlim_t K_*\big(\mathcal{M}^{p}(R/tR)\big),$$

where t runs over regular elements of R.

Thus the vanishing problem is reduced to the following concrete statement: given a regular element $t \in R$, one must find an element $f \in R$ avoiding all points of S such that the exact functor

$$\mathcal{M}^p(R/tR) \longrightarrow \mathcal{M}^p(R_f), \qquad M \longmapsto M_f,$$

induces the zero map on K-groups. Once this is done for every regular t, the filtered-colimit description shows that the original inclusion

$$\mathcal{M}^{p+1}(A) \hookrightarrow \mathcal{M}^p(A)$$

induces zero on K-theory.

This is the decisive reduction: the theorem is transformed from a statement about all codimension-$(p+1)$ supports at once into a statement about the behavior of codimension-p modules under passage from a hypersurface R/tR to a suitable open subset of $\mathrm{Spec}R$.

13.9. **Step 3: Quillen's normalization lemma.** The geometric input is the following variant of Noether normalization, which is precisely Lemma 5.12 in Quillen's paper.

> Let R be a smooth finite type algebra of dimension r over a field k, let $t \in R$ be a regular element, and let $S \subseteq \mathrm{Spec} R$ be finite. Then there exist algebraically independent elements $x_1, \dots, x_{r-1} \in R$ such that for
> $$B = k[x_1, \dots, x_{r-1}] \subseteq R$$
> one has:
> (i) R/tR is finite over B;
> (ii) R is smooth over B at the points of S.

Let us explain the proof. For each prime in S, choose a maximal ideal containing it; enlarging S if necessary, we may suppose that S itself consists of maximal ideals. Since R is smooth of dimension r over k, the module of Kähler differentials $\Omega^1_{R/k}$ is a projective R-module of rank r. Let
$$J = \bigcap_{\mathfrak{m} \in S} \mathfrak{m}.$$
Because $R/J^n \cong \prod_{\mathfrak{m} \in S} R/\mathfrak{m}^n$ is finite-dimensional over k for every n, one can choose a finite-dimensional k-subspace $V \subseteq R$ with two properties:

(a) V generates R as a k-algebra;
(b) for each $\mathfrak{m} \in S$ there exist elements $y_1, \dots, y_r \in V$ whose differentials form a basis of $\Omega^1_{R/k} \otimes_R k(\mathfrak{m})$ and which vanish at all other points of S.

The second property is obtained by choosing elements separately in the factors of the finite-dimensional algebra R/J^n and lifting them to R.

Now filter R/tR by degree in the elements of V: let $F_n(R/tR)$ be the k-subspace spanned by monomials of degree at most n in elements of V. The associated graded ring $\mathrm{gr}(R/tR)$ has Krull dimension $r-1$. Geometrically, one views
$$\mathrm{Proj}\Big(\bigoplus_{n \geq 0} F_n(R/tR)\Big)$$
as the projective closure of $\mathrm{Spec}(R/tR)$ inside the affine space determined by V; the part at infinity is $\mathrm{Proj}(\mathrm{gr}(R/tR))$, so its dimension is one less than that of R/tR, namely $r-2$. Hence $\mathrm{gr}(R/tR)$ has dimension $r-1$.

Choose a homogeneous system of parameters
$$z_1, \dots, z_{r-1} \in \mathrm{gr}(R/tR)$$
with each z_i of positive degree. Then $\mathrm{gr}(R/tR)$ is finite over the polynomial ring $k[z_1, \dots, z_{r-1}]$. Lift each z_i to an element $x'_i \in R$. Standard filtered-module arguments show that if the leading terms of elements generate the associated graded ring integrally, then the original elements generate the filtered ring integrally; therefore R/tR is finite over $k[x'_1, \dots, x'_{r-1}]$.

We still need smoothness at the points of S. By construction of V, for each i we may choose $v_i \in V$ so that the adjusted elements
$$x_i = x'_i + v_i, \qquad 1 \leq i \leq r-1,$$
have differentials $dx_1, \dots, dx_{r-1}$ linearly independent in $\Omega^1_{R/k} \otimes_R k(\mathfrak{m})$ for every $\mathfrak{m} \in S$. Put
$$B = k[x_1, \dots, x_{r-1}].$$
Because the differentials are independent, the Jacobian criterion implies that R is smooth over B at the points of S. On the other hand, replacing x'_i by $x'_i + v_i$ does not change

the leading term in the filtration, so the finiteness of R/tR over B is preserved. This proves the lemma.

The importance of the lemma is conceptual as well as technical: it rewrites the hypersurface R/tR as finite over an affine base of dimension $r-1$, while keeping R smooth of relative dimension one over that base near the points we care about.

13.10. **Step 4: the deformation square.** With the normalization lemma in hand, define

$$B' = R/tR, \qquad R' = R \otimes_B B'.$$

Because $B' \cong R/tR$ is finite over B, base change shows that R' is finite over R. There is a commutative square

$$\begin{tikzcd} R' \arrow[r, "u"] \arrow[d, "u'"'] & R \arrow[d] \\ B' \arrow[r] & B \end{tikzcd}$$

whose horizontal arrows are finite. Let $S' \subseteq \mathrm{Spec}R'$ be the finite set of points lying over S.

Since R is smooth over B at the points of S and $\dim R - \dim B = 1$, the morphism

$$\mathrm{Spec}R \longrightarrow \mathrm{Spec}B$$

is smooth of relative dimension one near S. Smoothness is stable under base change, so the morphism

$$\mathrm{Spec}R' \longrightarrow \mathrm{Spec}B'$$

is smooth of relative dimension one near S'. The diagonal section of a smooth morphism of relative dimension one is then a regular immersion of codimension one. In algebraic terms, if

$$I = \ker(R' \to B')$$

then I is locally free of rank one near the points of S', hence principal in a neighborhood of S'.

Because R' is finite over R, the inverse image of a neighborhood of S in $\mathrm{Spec}R$ contains a neighborhood of S' in $\mathrm{Spec}R'$. We may therefore choose an element $f \in R$ avoiding all points of S such that after replacing R by R_f the following properties hold simultaneously:

(a) the ideal I_f is principal on R'_f;
(b) R'_f is smooth over B', hence flat over B'.

If I_f is generated by an element $g \in R'_f$, multiplication by g gives an R'_f-linear isomorphism

$$R'_f \xrightarrow{\sim} I_f.$$

In particular, forgetting to R_f-modules, one has

$$I_f \cong R'_f.$$

The point of the construction is that the hypersurface $t = 0$ is now realized as the zero section inside a smooth one-dimensional family over B'. This is exactly the setup needed to produce a short exact sequence of exact functors whose two left-hand terms are isomorphic.

13.11. **Step 5: the exact sequence of functors and the vanishing argument.** Let $M \in \mathcal{M}^p(B')$. Start from the short exact sequence of B'-algebras

$$0 \longrightarrow I_f \longrightarrow R'_f \longrightarrow B'_f \longrightarrow 0.$$

Because R'_f is flat over B', tensoring with M over B' preserves exactness and yields a short exact sequence of R_f-modules

$$0 \longrightarrow I_f \otimes_{B'} M \longrightarrow R'_f \otimes_{B'} M \longrightarrow B'_f \otimes_{B'} M \longrightarrow 0.$$

The last term is canonically M_f, so we obtain the functorial exact sequence

$$0 \longrightarrow I_f \otimes_{B'} M \longrightarrow R'_f \otimes_{B'} M \longrightarrow M_f \longrightarrow 0. \tag{1}$$

Since R'_f is flat over B', if M has support of codimension at least p on SpecB', then $R'_f \otimes_{B'} M$ has support of codimension at least p on SpecR'_f; viewed via the finite map $R_f \to R'_f$, it therefore belongs to $\mathcal{M}^p(R_f)$. The same is true for $I_f \otimes_{B'} M$ and for M_f. Hence (1) is a short exact sequence in the exact category $\mathcal{M}^p(R_f)$, natural in M.

Thus we have an exact sequence of exact functors

$$0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow F_3 \longrightarrow 0$$

from $\mathcal{M}^p(B')$ to $\mathcal{M}^p(R_f)$, where

$$F_1(M) = I_f \otimes_{B'} M, \qquad F_2(M) = R'_f \otimes_{B'} M, \qquad F_3(M) = M_f.$$

By Step 4 there is an isomorphism $I_f \cong R'_f$ as R_f-modules. Tensoring with M gives a natural isomorphism of exact functors

$$F_1 \cong F_2.$$

Applying Quillen's additivity theorem, we obtain on every higher K-group the relation

$$(F_3)_* = (F_2)_* - (F_1)_*.$$

Since $(F_1)_* = (F_2)_*$, it follows that $(F_3)_* = 0$. Therefore the localization functor

$$\mathcal{M}^p(B') \longrightarrow \mathcal{M}^p(R_f), \qquad M \longmapsto M_f,$$

induces the zero map on all K-groups.

This is exactly the statement needed at the end of Step 2. Therefore for every regular element $t \in R$ one can find an f avoiding all points of S such that the functor

$$\mathcal{M}^p(R/tR) \longrightarrow \mathcal{M}^p(R_f)$$

induces zero on K-groups. Passing back through the filtered-colimit argument shows that the maps

$$K_*\big(\mathcal{M}^{p+1}(A)\big) \longrightarrow K_*\big(\mathcal{M}^p(A)\big)$$

vanish for every p. By Proposition 5.6, this is equivalent to the Gersten condition for A.

Theorem 13.7 (Gersten in Quillen's equicharacteristic cases)**.** *If A is a regular semi-local ring obtained from a smooth finite type algebra over a field by localizing at finitely many regular points, then the Gersten complex attached to the support spectral sequence of* Spec$A$ *is exact.*

Proof. Step 1 reduces the theorem to the case where R is smooth of finite type over a perfect field and $A = S^{-1}R$. In this situation Step 2 shows that it is enough to prove the following auxiliary assertion: for every regular element $t \in R$ and every $p \geq 0$, there exists an element $f \in R$ avoiding the primes of S such that the exact functor

$$\mathcal{M}^p(R/tR) \longrightarrow \mathcal{M}^p(R_f), \qquad M \longmapsto M_f,$$

induces the zero map on K-groups.

Fix such a regular element t. By Step 3, there are algebraically independent elements $x_1, \dots, x_{r-1} \in R$ and a polynomial subring $B = k[x_1, \dots, x_{r-1}] \subseteq R$ such that R/tR is finite over B and R is smooth over B at the points of S. Put $B' = R/tR$ and $R' = R \otimes_B B'$. Step 4 produces, after localizing at a suitable f not vanishing on S, a commutative square

$$\begin{tikzcd} R'_f \arrow[r] \arrow[d] & R_f \arrow[d] \\ B' \arrow[r] & B \end{tikzcd}$$

with R'_f flat over B' and with the kernel

$$I_f = \ker(R'_f \to B')$$

isomorphic to R'_f as an R_f-module.

For $M \in \mathcal{M}^p(B')$, Step 5 gives a functorial short exact sequence in $\mathcal{M}^p(R_f)$,

$$0 \longrightarrow I_f \otimes_{B'} M \longrightarrow R'_f \otimes_{B'} M \longrightarrow M_f \longrightarrow 0,$$

and the first two tensor functors are naturally isomorphic. Quillen's additivity theorem therefore implies that the third functor $M \mapsto M_f$ acts trivially on all K-groups. This proves the auxiliary assertion.

Since the auxiliary assertion holds for every regular element t, the filtered-colimit description of Step 2 shows that the inclusion

$$\mathcal{M}^{p+1}(A) \hookrightarrow \mathcal{M}^p(A)$$

induces the zero map on K-groups for every p. By Proposition 5.6, this is equivalent to the Gersten condition for $\operatorname{Spec} A$. Hence the Gersten complex attached to the support spectral sequence is exact, as claimed. □

13.12. **The Gersten complex as a flasque resolution.** Assume now that the Gersten condition holds stalkwise on a regular scheme X. Let $\mathcal{K}_n$ denote the Zariski sheaf associated with the presheaf

$$U \longmapsto K_n(U).$$

Sheafifying the Gersten complex coming from the support spectral sequence yields a complex of flasque sheaves

$$0 \to \mathcal{K}_n \to \bigoplus_{x \in X^{(0)}} (i_x)_* K_n(k(x)) \xrightarrow{d_1} \bigoplus_{x \in X^{(1)}} (i_x)_* K_{n-1}(k(x)) \xrightarrow{d_1} \cdots,$$

where $i_x \colon \operatorname{Spec} k(x) \to X$ is the canonical map.

Proposition 13.8 (Gersten resolution)**.** *If every local ring of X satisfies the Gersten condition, then the sheafified Gersten complex above is a flasque resolution of $\mathcal{K}_n$. In particular,*

$$H^p(X, \mathcal{K}_n) \cong H^p\Big(0 \to \bigoplus_{x \in X^{(0)}} K_n(k(x)) \to \bigoplus_{x \in X^{(1)}} K_{n-1}(k(x)) \to \cdots\Big).$$

Proof. Consider the presheaf complex obtained by taking the Gersten complexes of all open subsets of X:

$$0 \to U \mapsto K_n(U) \to U \mapsto \bigoplus_{x \in U^{(0)}} K_n(k(x)) \to U \mapsto \bigoplus_{x \in U^{(1)}} K_{n-1}(k(x)) \to \cdots.$$

Sheafifying term by term gives the complex displayed in the statement,

$$0 \to \mathcal{K}_n \to \bigoplus_{x\in X^{(0)}} (i_x)_* K_n(k(x)) \xrightarrow{d_1} \bigoplus_{x\in X^{(1)}} (i_x)_* K_{n-1}(k(x)) \to \cdots .$$

We must show that this is an exact flasque resolution.

First, each term on the right is flasque. Indeed, for a point $x \in X$ and an abelian group A, the skyscraper sheaf $(i_x)_*A$ has sections

$$\Gamma(U, (i_x)_*A) = \begin{cases} A, & x \in U, \\ 0, & x \notin U, \end{cases}$$

so every restriction map is either the identity or the zero map between equal source and target, hence surjective. A direct sum of such sheaves is therefore again flasque.

Second, exactness may be checked on stalks. Let $y \in X$. The stalk of the displayed sheafified Gersten complex at y is obtained by taking the filtered colimit over open neighborhoods $U \ni y$. Because

$$\mathrm{Spec}\mathcal{O}_{X,y} \cong \varprojlim_{y\in U} U$$

and because Quillen's support spectral sequence is compatible with these projective limits of affine neighborhoods, the stalk of the sheafified Gersten complex identifies with the Gersten complex attached to the local scheme $\mathrm{Spec}\mathcal{O}_{X,y}$. By hypothesis every local ring of X satisfies the Gersten condition, so that local Gersten complex is exact. Hence every stalk of that sheafified complex is exact.

A complex of sheaves is exact if and only if all of its stalks are exact. Therefore the displayed sheafified complex is an exact complex of flasque sheaves beginning with $\mathcal{K}_n$. It is thus a flasque resolution of $\mathcal{K}_n$.

Finally, sheaf cohomology may be computed from any flasque resolution. Taking global sections of that complex therefore yields

$$H^p(X, \mathcal{K}_n) \cong H^p\Big(0 \to \bigoplus_{x\in X^{(0)}} K_n(k(x)) \to \bigoplus_{x\in X^{(1)}} K_{n-1}(k(x)) \to \cdots\Big),$$

which is the asserted description. □

This is Proposition 5.8 in Quillen's paper and it is the sheaf-theoretic bridge from local exactness to global cohomology.

13.13. **Bloch's formula.** Take $n = p$. On the E_1-page of the support spectral sequence, the term in bidegree $(p, -p)$ is

$$E_1^{p,-p} = \bigoplus_{x\in X^{(p)}} K_0(k(x)) \cong \bigoplus_{x\in X^{(p)}} \mathbb{Z},$$

which is the free abelian group on codimension-p points, i.e. the group of codimension-p cycles. The first differential

$$d_1 \colon \bigoplus_{y\in X^{(p-1)}} K_1(k(y)) \longrightarrow \bigoplus_{x\in X^{(p)}} K_0(k(x))$$

is precisely the divisor map. Quillen proves that its image is the subgroup of cycles linearly equivalent to zero. Therefore its cokernel is the Chow group $CH^p(X)$.

Combining this identification with the Gersten resolution gives Bloch's formula.

Theorem 13.9 (Bloch's formula in Quillen's range)**.** *Let X be a regular scheme of finite type over a field. Then for every $p \geq 0$ there is a canonical isomorphism*

$$CH^p(X) \cong H^p(X, \mathcal{K}_p).$$

Proof. Apply the Gersten resolution of the previous proposition to the sheaf $\mathcal{K}_p$. The p-th cohomology of that flasque resolution is the cohomology at the term indexed by codimension p:

$$H^p(X, \mathcal{K}_p) \cong \frac{\ker\Big(\bigoplus_{x \in X^{(p)}} K_0(k(x)) \xrightarrow{d_1} \bigoplus_{x \in X^{(p+1)}} K_{-1}(k(x))\Big)}{\operatorname{im}\Big(\bigoplus_{y \in X^{(p-1)}} K_1(k(y)) \xrightarrow{d_1} \bigoplus_{x \in X^{(p)}} K_0(k(x))\Big)}.$$

For a field F, one has $K_0(F) \cong \mathbb{Z}$ and $K_{-1}(F) = 0$. Therefore the kernel in the numerator of the displayed quotient is simply

$$\bigoplus_{x \in X^{(p)}} K_0(k(x)) \cong \bigoplus_{x \in X^{(p)}} \mathbb{Z},$$

which is the free abelian group on codimension-p points of X, namely the group $Z^p(X)$ of codimension-p cycles.

It remains to identify the denominator. Quillen proves in Proposition 5.14 that the image of

$$d_1 \colon \bigoplus_{y \in X^{(p-1)}} K_1(k(y)) \longrightarrow \bigoplus_{x \in X^{(p)}} K_0(k(x))$$

is exactly the subgroup of cycles linearly equivalent to zero. The reason is the following. For each codimension-$(p-1)$ point y, one has

$$K_1(k(y)) \cong k(y)^\times,$$

and an element $f \in k(y)^\times$ defines a principal divisor on the closure $Y = \overline{\{y\}}$. Quillen compares the component of d_1 from $k(y)^\times$ to a codimension-p point $x \in Y$ with the discrete valuation map

$$\operatorname{ord}_{y,x} \colon k(y)^\times \longrightarrow \mathbb{Z},$$

ultimately reducing the comparison to the one-dimensional local case (his Lemma 5.16). Thus d_1 is exactly the divisor map in codimension p.

Consequently the quotient above is

$$\frac{Z^p(X)}{\{\text{cycles linearly equivalent to } 0\}} = CH^p(X).$$

This identifies $H^p(X, \mathcal{K}_p)$ canonically with the Chow group $CH^p(X)$, proving Bloch's formula. □

Remark 13.10. For $p = 1$ the formula reduces to $\operatorname{Pic}(X) \cong H^1(X, \mathcal{O}_X^\times) \cong H^1(X, \mathcal{K}_1)$, since $\mathcal{K}_1 \simeq \mathcal{O}_X^\times$ on regular schemes. For $p = 0$ one recovers $H^0(X, \mathbb{Z})$ and the free abelian group on connected components. The novelty begins in codimension ≥ 2, where higher K-theory encodes codimension-p algebraic cycles cohomologically.

13.14. **Why this is a genuine higher-dimensional payoff.** The support spectral sequence and Gersten's conjecture show that higher algebraic K-theory is not merely a stable linear invariant of rings. It organizes the codimension-by-codimension structure of a scheme. The residue fields $k(x)$ appear in exactly the way dictated by geometry, the first differential is a generalized divisor map, and the whole filtration is governed by localization. This explains why later developments in motivic cohomology, cycle modules, and coniveau filtrations all pass through Quillen's formalism.

In summary, Quillen's foundational paper already contains three of the most important geometric ideas in modern algebraic K-theory:

1. the support filtration and its exact couple;
2. the Gersten complex and its local exactness problem;
3. the identification of cycle groups with sheaf cohomology of K-theory sheaves.

That package is the real reason the higher theory immediately became central in algebraic geometry.

Appendix A. A brief post-Quillen outlook

After Quillen, the subject was expanded in several decisive directions. Waldhausen generalized higher K-theory to categories with cofibrations and weak equivalences, making it applicable to spaces and chain complexes. Thomason and Trobaugh globalized localization and descent in a far more flexible scheme-theoretic setting. Bass' negative K-groups were incorporated into nonconnective spectra, and later developments connected algebraic K-theory with cyclic homology, topological cyclic homology, motivic filtrations, and trace methods. But all these later theories still inherit Quillen's central principle: higher K-theory is a homotopy-theoretic invariant extracted from exact or exact-like categorical data.

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